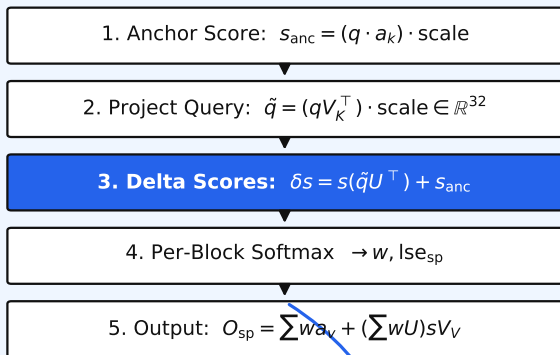


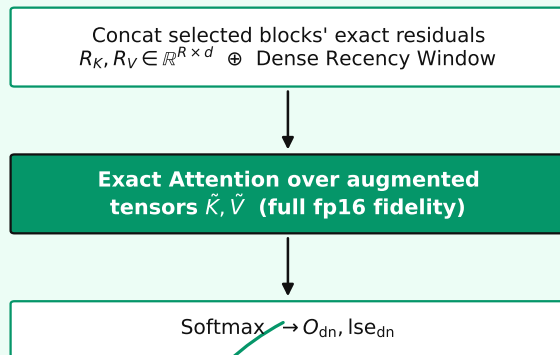
exact $q \cdot k$ over anchor + residual keys

Low-Rank Branch (Selected Blocks)

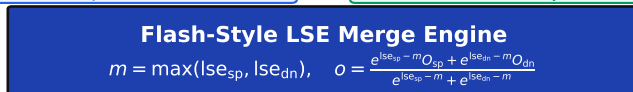


$O(K \cdot r \cdot B)$ compute — never materializes dense $\hat{K}$

Exact Branch (Residuals + Recency)



verbatim recall of needle tokens



NaN / inf-guarded: empty compressed set contributes zero weight